THREATREADY

User Manual

Data Security Operations & Simulation Lab

Threatready lets you practise real data-security work on a fictional company, Kestrel Ridge Financial. This manual shows you where everything is and walks you through your first simulation, one step at a time. Nothing you do here touches a real system or real people.

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All data in Threatready is synthetic.

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1 Getting started

You need a web browser and the address your administrator gave you (on a local install it is **http://127.0.0.1:5180**).

1. Open the address in Chrome or Edge.
2. Click **Create learner account** the first time. Next time, use **Sign in**.
3. Enter a **username**. Your email address works.
4. Enter a **display name**. This is the name shown on screen.
5. Choose a **password** of at least 10 characters, with an uppercase letter, a lowercase letter and a digit.
6. Click **Create account**. You land on the **Overview Dashboard**.

If the page says *Cannot reach the Threatready server*, the backend is not running. See **Troubleshooting** at the end.

2 The screen at a glance

Every page has the same areas:

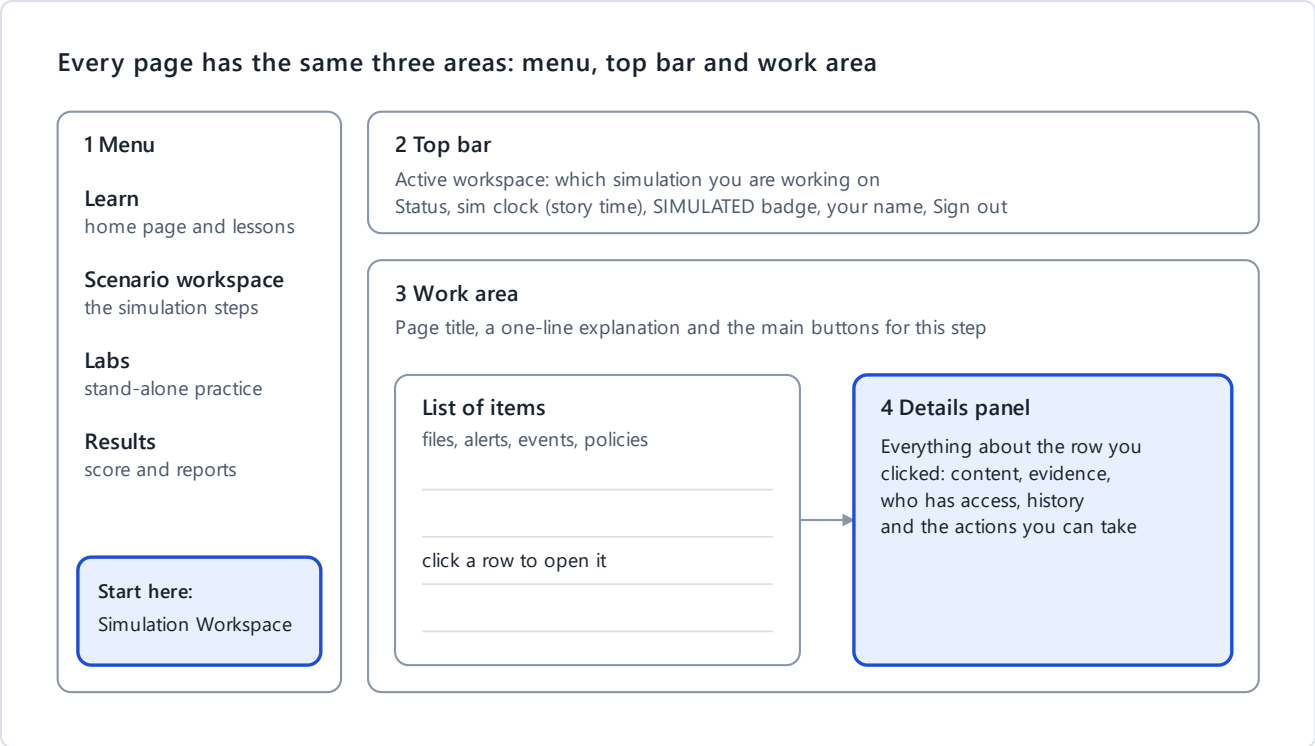


Figure 1. Threatready screen layout

Whatever page you are on, you choose *where to go* in the menu (1), check *which simulation you are in* on the top bar (2), and do the work in the middle (3). On most pages, clicking a row in a list opens its details on the right (4).

The top bar shows the same items everywhere:

Item on the top bar	What it means	Do I need to touch it?
Active workspace box	The simulation you are working on. Every page under *Scenario workspace* shows this simulation.	Only to switch between simulations
ACTIVE (or PAUSED, SUBMITTED)	Whether you can still make changes.	No
Sim clock	The date and time inside the story. It moves forward as you act. It is not your real clock.	No
SIMULATED	A reminder that everything here is made-up practice data.	No
Your name and **Sign out**	Who is signed in.	When you finish

3 The menu explained

The left menu has four groups. As a new user you only need two items at first: **Overview Dashboard** and **Simulation Workspace**. Everything under *Scenario workspace* works on the simulation you have open.

Menu item	What it is for	When you use it
Learn		
Overview Dashboard	Your home page: progress, next recommended task, recent work	Every time you sign in
Learning Paths	Lessons, levels L0–L5, knowledge checks and specialist pathways	To pick what to learn next
Scenario workspace		
Simulation Workspace	Start, pause, reset and run a simulation; shows your task checklist	Start here for every simulation
Data Discovery	Scan the company's files and databases for sensitive data	Step 1 of a simulation
Classification Studio	Label data (Public to Restricted), set owners, review who has access	Step 2
DLP Policy Studio	Write rules that allow, warn about or block data transfers, and test them	Step 3
Security Event Explorer	Search every event the simulation recorded	While investigating
Incident Investigation	Triage alerts, open a case, record evidence, contain and write the report	Steps 5–7
Data Security Engineering	Measure how well your rules work against test cases	Optional, advanced
Labs		
Encryption Lab, Database Security Lab, Privacy & Governance, Cloud Data Security / DSPM, AI Data Security	Stand-alone practice exercises (see **Practice labs**)	Any time; they do not need a simulation
Real-Tool Practice	Which real tools are available and step-by-step guides for outside tools	When you want to go beyond the platform

Menu item	What it is for	When you use it
Results		
Assessment & Reports	Submit your simulation, see your score, download reports	At the end of a simulation
Administrator / Lab Settings	Users, roles and audit log (administrators and instructors only)	Not needed by learners

The small codes next to some items (M01, M02 …) are the module numbers used in **Learning Paths**.

4 Your first simulation, step by step

The simulation *Unauthorized Customer-Data Export* takes about 75 minutes. You play the company's Data Protection Analyst: someone will try to move a customer file out of the company during a normal business day, and your job is to stop, investigate and fix it. The **Progress** checklist on the right of the Simulation Workspace ticks off each task as you complete it.

1. **Start the simulation.** Click **Simulation Workspace**, then **Start guided practice**. Guided mode gives hints; *independent assessment* gives none and picks a random version of the story. Read the **Briefing** card.
2. **Find the data.** Click **Data Discovery**, then **Run discovery scan**. Click the row `customer_master_export_2026Q1.csv`. On the right you see what it contains (card numbers, emails), that it has **no owner** and **no label**, and who can open it.
3. **Label the data.** Click **Classification Studio**, click the export row, choose a **Label** (think: bulk customer data with card numbers), choose a **Data owner**, write a short **Justification**, and click **Save classification**. Label the other unlabeled files too.
4. **Review access.** In Classification Studio, open the **Ownership & access review** tab. For each permission on the export, type a note and click **KEEP** or **REVOKE**. Revoking changes who can read the file.
5. **Write a rule.** Click **DLP Policy Studio**, then **New policy**. Give it a key such as `POL-DLP-CUSTOMER`, set **Outcome** to **BLOCK**, keep the condition *destination.sanctioned eq false*, click **+ Condition** to add *content.label label_at_least confidential*, and click **Save policy**.
6. **Test the rule.** Open the **Test workspace** tab and click **Test against saved policies**. Try one transfer that should be blocked and one that should be allowed (for example the marketing brochure). The trace shows exactly which condition matched.
7. **Run the business day.** Back in **Simulation Workspace**, click **Run business day**. Ten activities happen through the controls you set up. A message tells you how many events, alerts and blocked actions there were.
8. **Triage the alerts.** Click **Incident Investigation**. Click each alert, read *Why the rule fired* and the related events, choose a **Disposition** (real problem, legitimate activity, or not sure) and give a reason.
9. **Investigate.** On the most serious alert click **Open case from this alert**. In the case: click **Link as evidence** on the key events, add notes (an *Observation* must cite evidence; add an *Inference* and an *Unknown* too), then on **Scope & severity** record what was affected, use the **Risk model calculator**, and pick root causes.
10. **Contain and fix.** On **Containment & remediation**, preserve evidence *before* any device action, then disable the account and escalate to the Privacy Office. Fix the root cause: revoke the excessive permission and make sure your DLP rule blocks.
11. **Prove the fix.** In **Simulation Workspace** click **Run control-validation replay**. The day runs again with a test copy of the person's account; the risky transfer should now be stopped while normal work still succeeds.

12. **Write the report.** In the case, open the **Report** tab and fill in summary, root cause, residual risk and lessons learned. Click **Save report sections**.
13. **Submit.** Click **Assessment & Reports**, then **Submit for assessment**. You get a score out of 100 with an explanation for every point lost, and you can view or download the incident report.

To try again, click **Reset** in the Simulation Workspace. Your submitted result is kept and you can get a different version of the story.

5 Practice labs

Labs are short, self-contained exercises that do not need a simulation. Each lab page lists numbered exercises; a green **Exercise passed** badge appears when the server has checked your work. Every lab has a **Reset** button, so you can repeat it.

Lab	What you practise	Real or simulated	Exercises
Encryption Lab	Spotting a tampered file with hashes, encrypting a file and proving a wrong password or tampering is rejected, rotating encryption keys, what TLS hides, fixing a weak encryption setup	Real (exercises 1–3), simulated (4–5)	5
Database Security Lab	Least-privilege roles, SQL injection vs safe queries, masking card numbers, spotting suspicious queries in an audit log, backup and restore	Real (a private practice database)	5
Privacy & Governance	Deciding which fields a marketing list really needs, retention rules, deleting old records without touching those under legal hold, a findings report	Simulated	3
Cloud Data Security / DSPM	Finding the most exposed storage bucket, reducing exposure without breaking the public website	Simulated	2
AI Data Security	Stopping sensitive prompts to an unapproved AI tool, stopping an AI assistant from showing documents a user should not see, safe AI log settings	Simulated	3

Real means the operation actually runs on your computer, on throw-away practice files. *Simulated* means the platform models the situation; no real cloud account or AI service is contacted.

6 Lessons, levels and your results

A lesson counts as complete only when you have done the practical work it asks for: a simulation task, a lab exercise or a short quiz. Reading alone does not complete a lesson.

- **Levels** run from L0 Foundation to L5 Independent Mastery. A level unlocks when you finish enough lessons in the level below.
- **Pathways** let you focus: Foundations, DLP Engineer, SOC Analyst or Data Protection Engineer. Choose one in **Learning Paths**; its lessons unlock even if they belong to a higher level.
- **Quizzes** have four questions and need 75% to pass. You see which answers were wrong, not the right answer, so you can retry fairly.

When you submit a simulation you get a **score out of 100**, made of five parts:

Part	Weight	What it checks
Technical correctness	25%	Correct labels and removing excessive access
Detection and analysis	25%	Alert decisions, the case conclusion, severity
Evidence quality	20%	Linked evidence and clear notes
Remediation and validation	20%	The replay shows the problem fixed without blocking normal work
Documentation and escalation	10%	Report sections and escalating to Privacy or Legal

You pass with **70 or more**, unless you hit a *critical failure*: wiping the person's laptop before preserving evidence, or never proving your fix with a replay. Every lost point comes with a one-line explanation and a suggested lesson.

Reports (in **Assessment & Reports**) are built from what you actually did: data inventory, classification register, policy tests, incident timeline, the full incident report, remediation results and your assessment. Download them as Markdown, HTML, JSON or CSV.

Scores are a platform result for this exercise. They are not a professional certification.

7 Words and badges you will see

Word or badge	Meaning
SIMULATED (purple)	Produced by the simulation with made-up data. Nothing real happened.
REAL LAB (green)	A real operation ran on your computer, on throw-away practice files.
ACTIVE / PAUSED / SUBMITTED	State of your simulation. Only an *active* simulation can be changed.
UNLABELED / PUBLIC / INTERNAL / CONFIDENTIAL / RESTRICTED	How sensitive a piece of data is, from not yet decided to most sensitive.
ALLOWED / BLOCKED / DENIED	ALLOWED: the action went through. BLOCKED: a rule stopped it. DENIED: the person had no permission or their account was disabled.
MONITOR / WARN / BLOCK	What a rule does when it matches: only record it, warn the user, or stop the action.
Alert	A message raised when a rule matches something worth a look.
True positive / Benign positive / False positive / Undetermined	Your decision on an alert: a real problem; correct match but legitimate work; the rule matched something it should not; the evidence cannot tell.
Case	Your investigation folder: linked evidence, notes, scope, actions and report.
Observation / Inference / Unknown	A note type: what the evidence shows; what you conclude from it; what you cannot tell.
Remediation	An action that changes the environment, such as disabling an account or removing a permission. It is checked right after you apply it (VERIFIED).
Control-validation replay	Re-running the day with a test copy of the person's account to prove your fix works.
Sim clock	The simulated date and time inside the story (UTC), not your real clock.
DLP	Data loss prevention: rules that stop sensitive data leaving the company.

8 Troubleshooting

Problem	What to do
Cannot reach the Threatready server	The backend is not running. Start it (see the README) or ask your administrator.
Invalid input when creating an account	Username: letters, digits, `. _ - + @` (an email works). Password: 10+ characters with upper- and lower-case letters and a digit.
Buttons are greyed out	The simulation is paused or submitted. Open **Simulation Workspace** and click **Resume**, or **Reset** to start a new attempt.
A page says *No active scenario workspace*	Start or pick a simulation in **Simulation Workspace**, or choose one in the **Active workspace** box at the top.
Run control-validation replay is greyed out	Run the business day once first.
No alerts appeared	Run the business day first. If you disabled the bulk-download rule, alerts for big downloads will not fire.
A remediation shows VERIFICATION FAILED	The change did not stick (for example the account was re-enabled). Click **Re-verify** after correcting it.
Too many login attempts	Wait one minute and try again.
I want to start over	**Reset** in Simulation Workspace keeps past results; each lab page has its own **Reset**.